

Pembrolizumab (Keytruda®)

Pembrolizumab (Keytruda[®]) is an immunotherapy drug used to treat many types of cancer.

On this page

- [What is pembrolizumab \(Keytruda®\)?](#)
- [How does pembrolizumab work?](#)
- [Having pembrolizumab](#)
- [About side effects](#)
- [Side effects while treatment is being given](#)
- [Very common side effects](#)
- [Other side effects](#)
- [Other important information](#)
- [Find the right information for you](#)
- [How we can help](#)

What is pembrolizumab (Keytruda®)?

Pembrolizumab[®] is a type of immunotherapy drug called a checkpoint inhibitor. Pembrolizumab is also called Keytruda[®]. It is used to treat many types of cancer.

It is best to read this information with our general information about [immunotherapy drugs](#) and the [type of cancer](#) you have.

Your cancer team will tell you more information about this treatment and the possible side effects before you agree ([consent](#)) to have treatment.

More information about this treatment

This information is correct at time of publishing. But sometimes the types of cancer this treatment is used for, or treatment side effects, may change between revision dates.

You can talk to your cancer team if you want more detailed information about this treatment. Or visit the [electronic Medicines Compendium \(eMC\)](#) website, which has patient information leaflets (PIL) for individual drugs.

How does pembrolizumab work?

The immune system protects the body against illness and infection. Immunotherapy drugs use the immune system to find and attack cancer cells.

Checkpoint inhibitors affect a type of white blood cell called a lymphocyte. Lymphocytes are an important part of the immune system. When they are active, they can attack other cells, including cancer cells. But if they receive a certain signal from the other cell, they switch off (become inactive) and do not attack it.

Checkpoint inhibitors block the signals that switch off lymphocytes. They do this by attaching to either the cancer cell or the lymphocyte. This means the lymphocyte stays active and can attack the cancer cell.

Before having treatment with pembrolizumab, your doctor may do a test to check the levels of a protein called PD-L1 on the surface of the cancer cells. This will depend on the cancer type.

Having pembrolizumab

You usually have pembrolizumab at a day unit or clinic as an outpatient. You may have it on its own or with other cancer drugs.

During a course of treatment, you will meet someone from your cancer team, such as a:

- cancer doctor
- specialist nurse
- specialist pharmacist.

This is who we mean when we mention doctor, nurse or pharmacist in this information.

During your course of treatment, you will have regular blood tests. This is to check that it is safe for you to have treatment.

You will meet with a doctor, nurse or pharmacist before you have treatment. They will talk to you about your blood results and ask how you have been feeling. If your blood results are okay, the pharmacy team will prepare your treatment.

You have pembrolizumab as an injection under the skin (subcutaneous). Or you may have pembrolizumab as a drip into a vein (intravenous infusion).

As an injection under your skin (subcutaneous)

You may have pembrolizumab as an injection under the skin (subcutaneous) in the tummy area (abdomen) or thigh. This takes 1-2 minutes.

As a drip into a vein (intravenous infusion)

Or you may have pembrolizumab as a drip into the vein (intravenous infusion). The drip is usually connected to a pump, which gives you the treatment over 30 minutes.

Your nurse will give you pembrolizumab through one of the following:

- a [cannula](#) – a short, thin tube the nurse puts into a vein in the arm or hand
- a [central line](#) – a fine tube that goes under the skin of the chest and into a vein close by
- a [PICC line](#) – a fine tube that is put into a vein in the arm and goes up into a vein in the chest
- an [implantable port](#) (portacath) – a disc that is put under the skin on the chest or arm and goes into a vein in the chest.

Your doctor may switch how you have treatment from having it as a drip into your bloodstream to having an injection under the skin. Sometimes, it may be changed from having it as an injection to having it as a drip.

Your cancer team may give you other medicines to take home to help with side effects. Take all your capsules or tablets exactly as they tell you to.

Your course of treatment

How often and for how long you have pembrolizumab depends on the type of cancer. You may have pembrolizumab once every 3 weeks, or once every 6 weeks. Your doctor, nurse or pharmacist will explain more about your treatment plan.

About side effects

We explain the most common side effects of this treatment here. We also include some that are less common. This treatment can make the immune system too active. This can cause side effects that may be serious if not treated quickly.

You may get some of the side effects we mention, but you are unlikely to get all of them. And you may have some side effects, including rarer ones, that we have not listed here.

Other cancer treatments may cause different side effects. If you are also having other cancer treatment, you may have other side effects.

Do not try to treat side effects yourself. Always tell your doctor, nurse or pharmacist about any side effects you have. They can give you:

- drugs to help control some side effects
- advice about managing side effects.

It is important to take any drugs exactly as explained. This means they will be more likely to work for you.

Serious and life-threatening side effects

Some cancer treatments can cause serious side effects. Sometimes, these may be life-threatening. Your doctor, nurse or pharmacist can explain the risk of these side effects to you.

Delayed side effects of immunotherapy drugs

Side effects can start during treatment but some may start weeks, months, or sometimes more than a year after you finish treatment.

If you have symptoms or side effects that start during treatment or after treatment ends, contact the hospital straight away on the 24-hour number. Side effects can become serious if they are not treated.

Contact the hospital

Your doctor, nurse or pharmacist will give you 24-hour contact numbers for the hospital. If you feel unwell or need advice, you can call at any time of the day or night. Save these numbers in your phone or keep them somewhere safe.

Your patient alert card

They will also give you a card with information about your treatment. It is important to keep the card with you during treatment and after treatment ends. You should show it to any doctor or healthcare professional caring for you.

Side effects while treatment is being given

Some people have side effects while they are having this treatment or shortly after they have it.

Allergic reaction

Some people have an allergic reaction while having this treatment. Signs of a reaction can include:

- feeling hot or flushed
- shivering
- itching
- a skin rash
- feeling dizzy or sick
- a headache
- feeling breathless or wheezy
- swelling of your face or mouth
- pain in your back, tummy or chest.

Your nurse will check you for signs of a reaction during your treatment. If you feel unwell or have any of these signs, tell them straight away. If you do have a reaction, they can treat it quickly.

Sometimes a reaction happens a few hours after treatment. If you develop any of these signs or feel unwell after you get home, contact the hospital straight away on the 24-hour number.

Always call 999 if swelling happens suddenly or you are struggling to breathe.

Injection site

If you have pembrolizumab as a subcutaneous injection, the skin in the area that was injected may become:

- painful
- swollen
- red, if you have white skin
- darker, if you have black or brown skin

If this happens, let your doctor know. Painkillers may help.

Very common side effects

These side effects happen to 10 or more people in every 100 people (10% or more) who have this treatment.

Diarrhoea or tummy pain

You may have [diarrhoea](#) or tummy (abdominal) pain. Sometimes it can cause serious inflammation of the bowel called colitis.

Contact the hospital straight away on the 24-hour number if you have any of the following symptoms during treatment, or after treatment ends:

- passing more stools (poo) than is normal for you
- passing stools at night
- if you have a stoma, it is more active or needs emptied more often than usual
- watery or loose stools
- uncomfortable tummy cramps or pain
- a temperature over 37.5°C
- blood or mucus (slime) in your stool.

Feeling tired (fatigue)

[Feeling tired](#) is a common side effect of this treatment. Try to pace yourself and plan your day so you have time to rest. Gentle exercise, like short walks, can help you feel less tired.

Sometimes, this treatment can affect glands that make hormones. This can make you feel tired. Contact the hospital on the 24-hour number if either of these things happen during treatment or after it ends:

- your tiredness gets worse or stops you doing your usual daily activities
- you feel weak or shaky.

If you feel sleepy, do not drive or use machinery.

Feeling sick

You may feel sick during this treatment, but this is usually mild. Your doctor, nurse or pharmacist may prescribe you anti-sickness drugs to help prevent or control sickness. Take the drugs exactly as they tell you to. [If you feel sick](#), take small sips of fluid often and eat small amounts regularly. It is important to drink enough fluids.

If you continue to feel sick, or if you are sick (vomit) 1 to 2 times in 24 hours, contact the hospital on the 24-hour number as soon as possible. They will give you advice.

Constipation

This treatment can cause [constipation](#). Constipation means that you are not able to pass stools (poo) as often as you normally do. It can become difficult or painful. Here are some tips that may help:

- Drink at least 2 litres (3½ pints) of fluids each day.
- [Eat high-fibre foods](#), such as fruit, vegetables and wholemeal bread.
- Do regular gentle exercise, like going for short walks.

If you have constipation, contact the hospital on the 24-hour number for advice. They can give you drugs called laxatives to help.

If you have not been able to pass stools for over 2 days and are being sick, contact the 24-hour number straight away.

Loss of appetite

This treatment can affect your appetite. Don't worry if you do not eat much for 1 or 2 days. But if your appetite does not come back after a few days, or if you are losing weight, tell your doctor, nurse or pharmacist. They can give you advice. They may give you food or drink supplements. Or they may suggest changes to your diet or eating habits to help.

Headaches

This treatment may cause headaches. If you have headaches, tell your doctor, nurse or pharmacist. They can give you advice about painkillers that may help. Tell them if the headache does not get better, or gets worse.

Skin changes

This treatment may cause inflammation in the skin. This is usually mild, but if it is not treated it can become serious. You may need creams, steroids or other medicines to treat skin problems.

Contact the hospital on the 24-hour number as soon as possible, if you notice any skin changes during treatment, or after treatment ends. This includes:

- a rash or bumps on the skin
- dry, tight or itchy skin
- areas that are a different colour to your usual skin tone.

Here are some things you can do to take care of your skin during treatment:

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Wash with lukewarm water and mild, unperfumed, soap-free cleansers.

- Moisturise your skin regularly with unperfumed moisturisers for dry skin conditions.
- Use sun cream of at least SPF 30 to protect your skin from the sun. Cover up with clothing and a hat.

Contact the hospital straight away on the 24-hour number if you have any of these serious skin symptoms:

- a rash that is spreading
- swollen, blistering or peeling skin
- flu-like symptoms, such as a high temperature and joint pain.

Muscle and joint pain

This treatment can cause sore or weak muscles. Your joints may also be stiff, sore or swollen. This is called arthritis. If you already have problems with joint pain, this may become worse.

Contact the hospital on the 24-hour number if you have any of these symptoms during treatment or after it ends:

- new or worsening aches or pains
- new or worsening stiffness in a joint
- swollen joints or muscle weakness.

Fluid build-up

This treatment can cause a build-up of fluid in the body. This will slowly get better after treatment ends. Contact the hospital on the 24-hour number if you:

- are gaining weight
- have swelling in your face, legs or ankles.

They can give you advice and treatment to help.

Hormone changes

This treatment may permanently affect glands in the body that produce hormones. Hormones control different processes in the body. You will have regular blood tests to check some hormone levels, such as your thyroid. You may need to take tablets daily to correct your hormone levels.

Contact the hospital on the 24-hour number if you have any of these side effects during treatment or after it ends:

- increased sweating
- problems with constipation
- weight gain or loss
- dizziness or fainting
- feeling or being sick

- feeling more hungry or thirsty than usual
- loss of sex drive
- passing urine (peeing) more often than usual
- headaches that do not go away
- feeling more tired than usual (fatigue)
- mood changes
- changes to your heartbeat (palpitations).

Effects on the lungs

This treatment can cause inflammation of the lungs. This is called pneumonitis. Contact the hospital straight away on the 24-hour number if you notice any of these changes during treatment or after it ends:

- breathlessness
- a cough that does not go away
- wheezing
- a fever, with a temperature over 37.5°C.

You should also tell them if any existing breathing problems get worse. You may have:

- tests to check your lungs
- treatment, if needed.

Anaemia (low number of red blood cells)

This treatment can reduce the number of red blood cells in your blood. Red blood cells carry oxygen around the body. If the number of red blood cells is low, this is called anaemia. You may feel:

- very low in energy
- breathless
- dizzy and light-headed.

If you have these symptoms, contact the hospital straight away on the 24-hour number. You may need treatment for anaemia. If you are very anaemic, you may need a drip to give you extra red blood cells. This is called a blood transfusion.

Other side effects

These side effects happen to less than 10 in 100 people (less than 10%) who have this treatment. Some of them are much rarer than this but they are still important to know about. Rare means a side effect that happens to less than 1 in 1,000 people (less than 0.1%).

Low number of white blood cells

This treatment can reduce the number of white cells in your blood. These cells fight infection. If you have any signs of an [infection](#), tell your doctor, nurse or pharmacist. These signs include:

- a cough
- a sore throat
- diarrhoea
- needing to pass urine (pee) often, or discomfort when you pass urine.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms:

- a temperature above 37.5°C
- a temperature below 36°C
- you feel unwell, even with a normal temperature.

Bruising and bleeding

This treatment can reduce the number of platelets in your blood. Platelets are cells that help the blood to clot.

If the number of platelets is low, you may bruise or bleed easily. You may have:

- nosebleeds
- bleeding gums
- heavy periods
- blood in your urine (pee) or stools (poo)
- tiny red, brown or purple spots that may look like a rash – these spots can be harder to see if you have black or brown skin.

If you have any unexplained bruising or bleeding, contact the hospital straight away on the 24-hour number. You may need a drip to give you extra platelets. This is called a platelet transfusion.

Dry mouth

You may get a [dry mouth](#). Drink plenty of fluids. Add sauces and gravies to your food to keep it moist. If you have dry lips, using a lip balm or Vaseline[®] can help.

Changes to your taste

Some foods may taste different or have no taste. Try different foods to find out what tastes best to you. You may also get a bitter or metallic taste in your mouth. Your doctor, nurse or pharmacist can give you advice. It might help to try:

- sucking sugar-free sour or boiled sweets
- eating cold foods
- eating sharp-tasting fresh fruit.

Taste changes usually get better after treatment ends. We have more information about [coping with changes to taste](#).

Numb or tingling hands or feet (peripheral neuropathy)

This treatment may affect the nerves in your fingers and toes. This can cause numbness, tingling or pain in your hands or feet. This is called [peripheral neuropathy](#). You might find it hard to do fiddly tasks such as fastening buttons or tying shoelaces.

If you have these symptoms, always tell your doctor, nurse or pharmacist. They sometimes need to change the drug or the dose of the drug. The symptoms usually improve slowly after treatment ends. But for some people they continue and are a long-term side effect of treatment.

Difficulty sleeping (insomnia)

Sometimes this treatment can affect your sleep. Tell your doctor, nurse or pharmacist if cancer treatment makes it [difficult to sleep](#).

Effects on the liver

This treatment may affect how your liver works. It can cause liver inflammation called hepatitis. You will have regular blood tests to check how your liver is working. Tell your doctor, nurse or pharmacist if you have loss of appetite.

Sometimes liver changes can be serious. Contact the hospital straight away on the 24-hour number if you have any of these symptoms during treatment or after it ends:

- yellow skin or eyes
- feeling very sleepy
- dark urine (pee)
- unexplained bleeding or bruising
- pain in the right side of your tummy (abdomen).

If you have black or brown skin, changes in your skin colour may be less noticeable. You might first notice the whites of your eyes look yellow, or that the colour of your urine has changed.

Effects on the kidneys

This treatment can affect how the kidneys work. It may cause kidney inflammation called nephritis. You will have regular blood tests to check how your kidneys are working.

Contact the hospital straight away on the 24-hour number during treatment, or after it ends, if you:

- have blood in your urine (pee)
- are passing less urine or peeing less often than usual.

Eyesight changes

This treatment can affect your eyes and eyesight. Contact the hospital straight away on the 24-hour number if you have any of these symptoms during treatment or after it ends:

- dry, itchy, painful, or swollen eyes

- eyes that are sensitive to light
- changes to your eyesight.

Effects on the heart

This treatment can affect [how the heart works](#). You may have [tests](#) to check how well your heart is working. These may be done before, during and after treatment.

Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- breathlessness
- dizziness
- changes to your heartbeat (palpitations)
- swollen feet and ankles.
- flu-like symptoms, such as body aches, pains and tiredness.

Other conditions can cause these symptoms, but it is important to get them checked by a doctor.

Always call **999** if you have:

- chest pain, pressure, heaviness, tightness or squeezing across the chest
- difficulty breathing.

Effects on the nervous system

Rarely, this treatment can affect the brain and nerves. These effects may be mild at first but can become serious if not treated. Symptoms may include:

- weakness, numbness or tingling in your arms, legs or face
- a headache that does not get better
- seeing double or having drooping eyelids
- difficulty turning or bending your neck
- drowsiness or confusion
- problems with moving or speaking
- fits (seizures).

If you have any of these symptoms during treatment or after it ends, it is important to either:

- contact the hospital straight away on the 24-hour number
- go to the hospital straight away
- call **999**.

Do not drive yourself to hospital.

Raised blood sugar levels

This treatment may raise your blood sugar levels and increase the risk of diabetes.

You will have regular blood tests to check this.

Contact the hospital on the 24-hour number if you have these symptoms during treatment or after it ends:

- feeling thirsty all the time
- needing to pass urine (pee) more often than usual
- feeling more tired than usual.
- losing weight without trying.

If you have diabetes, your blood sugar levels may be higher than usual. You may need advice and support from your GP or diabetes team about managing this.

Effects on the pancreas

This treatment can cause inflammation of the pancreas. This is called pancreatitis. You will have regular blood tests to check your pancreas.

Contact the hospital straight away on the 24-hour telephone number you have been given if you have any of these symptoms:

- severe pain in the centre of your tummy (abdomen)
- feeling or being sick (vomiting)
- indigestion
- a temperature above 37.5°C
- fast heartbeat (tachycardia) or rapid breathing
- yellow skin or eyes – this may be harder to notice if you have black or brown skin
- tenderness or swelling of the tummy.

High blood pressure

This treatment can cause high blood pressure. Tell your doctor or nurse if you have ever had any problems with your blood pressure. Your nurse will check it regularly during your treatment.

Other important information

Driving

Do not drive if you:

- feel dizzy
- have blurred vision
-

feel very tired.

If you are worried about this, talk to your doctor, nurse or pharmacist.

Other medicines

Some medicines can affect how this treatment works or be harmful while you are having it. Always tell your cancer doctor, nurse or pharmacist about any drugs you are taking or planning to take, such as:

- medicines you have been prescribed
- medicines you buy in a shop, pharmacy or online
- vitamins or supplements
- [herbal drugs and complementary or homeopathic therapies](#)
- recreational drugs – for example, cannabis.

Blood clot risk

Cancer and some cancer treatments can increase the risk of a [blood clot](#). Contact the hospital straight away on the 24-hour number if you have any of these symptoms during or after treatment:

- throbbing pain or swelling in a leg or arm
- reddening of the skin in the area – if you have black or brown skin, this can be harder to notice, but the skin might become darker
- suddenly feeling breathless or coughing.

Always call **999** if you have:

- chest pain
- difficulty breathing.

A blood clot is serious, but it can be treated with drugs called anticoagulants. These thin the blood. Your doctor, nurse or pharmacist can give you more information about preventing and treating blood clots.

Vaccinations

Cancer doctors usually recommend that people with cancer have vaccinations for flu and [coronavirus \(covid\)](#). They may also recommend other vaccines, such as Shingrix[®] for shingles. These all help reduce your risk of serious illness from these infections. Most people can have these vaccines, including people with weak immune systems.

You should not have live vaccines if your immune system is weak. This includes if you are having or recently had chemotherapy, radiotherapy or other cancer treatments that affect your immune system. Live vaccines can make you unwell because they contain a very weak version of the illness they protect you against. There are several live vaccines, including the yellow fever vaccine.

It is important to ask your doctor, nurse or pharmacist for advice about having vaccinations. They can explain what vaccines are right for you and when it is best to have them.

Contraception

Your doctor, nurse or pharmacist will advise you not to [get pregnant](#) or [make someone pregnant](#) while having this treatment and for some time afterwards. The drugs may harm a developing baby. It is important to use contraception to prevent pregnancy. Follow their advice about:

- what types of contraception to use
- how long after treatment you should continue to use contraception.

Breastfeeding

You are advised not to [breastfeed](#) while having this treatment, or for some time after treatment ends. This is because the drugs could be passed to the baby through breast milk.

Your doctor, nurse or pharmacist can give you more information.

Fertility

Some cancer drugs can affect whether you can [get pregnant](#) or [make someone pregnant](#). If you are worried about this, it is important to talk with your doctor before you start treatment.

Medical and dental treatment

If you need medical treatment for any reason other than cancer, always tell the healthcare professional that you are having cancer treatment. Show them your patient alert card. Give them the contact details for your cancer doctor or cancer team so they can ask for advice.

If you have appointments with a dentist or optician, always tell them you are having cancer treatment. Talk to your cancer team before you have any dental treatment.

Find the right information for you

A cancer diagnosis can feel overwhelming, with so much to process and understand. You don't have to do this alone. Answer a few quick questions and we'll show you the support that fits your situation.

[Start now](#)

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References



Visit the [electronic Medicines Compendium \(eMC\)](#) to download a Patient Information Leaflet (PIL) for more detailed information. The leaflet lists all known side effects.

[Cancer Information Development team](#)

Author

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Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

How we can help

How to print a PDF from this website

Instructions on how you can make a PDF version of this page using the print function on your browser.

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help